Hydraulic System Maintenance Report

Generated: 2026-04-09 22:43:38

Executive Summary

Total cycles analysed: 5

Cycles requiring attention: 4

Healthy cycles: 1

Overall system status: ACTION REQUIRED

System Health Overview

Pump condition:	**Severe leakage**
Accumulator condition:	**Optimal pressure**

Cycle 2 - Detailed Finding

Pump Leakage Probabilities:

No leakage	**2%**
Weak leakage	**92%**
Severe leakage	**6%**

Accumulator Condition Probabilities:

Close to failure	**0%**
Severely reduced pressure	**0%**
Slightly reduced pressure	**2%**
Optimal pressure	**98%**

Top Contributing Sensors:

[+] System Efficiency (Average) - high impact (value: 57.6486)

[+] System Efficiency (Variability) - high impact (value: 22.8378)

[+] Volume Flow 1 (Average) - high impact (value: 6.3483)

AI Maintenance Finding:

FINDING:

The hydraulic system is experiencing weak leakage, confirmed with a diagnosis of 92% confidence, which requires immediate attention.

SENSOR EVIDENCE:

System efficiency sensors indicate a high level of risk, showing an average efficiency of 57.6486, high variability of 22.8378, and a maximum efficiency of 76.585. The volume flow 1 sensor is also showing high risk, reading an average of 6.3483. The motor power sensor is showing a moderate risk level with an average of 2403.7806, but it's worth noting that these sensors indicate potential issues that may lead to increased energy consumption and efficiency losses.

CONSEQUENCE OF DEFERRAL:

If this leakage issue is not addressed, it may cause premature wear on system components, leading to downtime and increased maintenance costs.

RECOMMENDED ACTION:
I recommend the shift engineer visually inspect the pump for signs of leakage and replace any worn or damaged components immediately to prevent further damage.

Cycle 3 - Detailed Finding

Pump Leakage Probabilities:

No leakage	**0%**
Weak leakage	**98%**
Severe leakage	**2%**

Accumulator Condition Probabilities:

Close to failure	**0%**
Severely reduced pressure	**0%**
Slightly reduced pressure	**1%**
Optimal pressure	**99%**

Top Contributing Sensors:

[+] System Efficiency (Average) - high impact (value: 56.6581)

[+] System Efficiency (Variability) - high impact (value: 23.3455)

[+] Volume Flow 1 (Average) - high impact (value: 6.3091)

AI Maintenance Finding:

FINDING:
A weak leakage was detected in the pump, indicating a potential loss of hydraulic fluid, and this situation is classified as a CONFIRMED diagnosis with a severity level of Weak leakage.

SENSOR EVIDENCE:
The pump assessment flagged system efficiency, with an average of 56.6581 indicating that the pump is operating at a lower efficiency than expected. This means the pump is not pushing the hydraulic fluid at its optimal rate, potentially leading to the detected weak leakage. In addition, the volume flow 1 sensor showed an average of 6.3091, indicating that the flow rate is lower than expected.

CONSEQUENCE OF DEFERRAL:
If this issue is not addressed, the pump's performance will continue to degrade, potentially causing further losses of hydraulic fluid, equipment malfunction, and potentially leading to costly repairs or even equipment failure.

RECOMMENDED ACTION:
Immediately inspect the pump for potential signs of leakage and perform necessary repairs to prevent any further fluid

loss and ensure optimal pump performance.

Cycle 4 - Detailed Finding

Pump Leakage Probabilities:

No leakage	1%
Weak leakage	14%
Severe leakage	86%

Accumulator Condition Probabilities:

Close to failure	2%
Severely reduced pressure	0%
Slightly reduced pressure	1%
Optimal pressure	98%

Top Contributing Sensors:

[+] System Efficiency (Variability) - high impact (value: 22.5673)

[+] Volume Flow 1 (Average) - high impact (value: 6.3054)

[+] Volume Flow 1 (Variability) - high impact (value: 2.8525)

AI Maintenance Finding:

Finding:
Severe leakage has been detected in the hydraulic system.

Sensor Evidence:
The system efficiency sensor is showing high variability, which means that the system's operation is not consistent and is likely causing the leakage. The volume flow sensor is also indicating high variability and an average flow rate that's higher than expected, which further supports the presence of leakage. Additionally, the system pressure sensor is showing an average pressure that's below what's considered normal.

Consequence of Deferment:
If this issue is not addressed, the severe leakage will continue to cause the hydraulic fluid to be wasted, potentially leading to system failure and downtime.

Recommended Action:
Perform a system inspection and repair or replace the pump with a spare unit as soon as possible, due to the severe leakage observed.

Cycle 5 - Detailed Finding

Pump Leakage Probabilities:

Hydraulic System Maintenance Report

Generated: 2026-04-09 22:44:05

No leakage	4%
Weak leakage	2%
Severe leakage	94%

Accumulator Condition Probabilities:

Close to failure	2%
Severely reduced pressure	1%
Slightly reduced pressure	11%
Optimal pressure	86%

Top Contributing Sensors:

[+] Volume Flow 1 (Average) - high impact (value: 3.0191)

[+] System Efficiency (Average) - high impact (value: 27.1264)

[+] Motor Power (Average) - high impact (value: 2676.6586)

AI Maintenance Finding:

FINDING:
Severe leakage in the pump system has been detected.

SENSOR EVIDENCE:
The volume flow sensor has detected an unusual increase in flow rate, indicating potential bypassing of pressurized fluid, which contributes to severe leakage. The system efficiency sensors have also detected high and moderate level increases in pressure and flow, suggesting inefficient pump performance. The high reading from the motor power sensor may imply additional strain on the motor due to this inefficient pump behavior.

CONSEQUENCE OF DEFERRAL:
If severe leakage is not addressed, it may cause a significant reduction in system performance, potential failure of connected components, and potential environmental hazards from the spilled fluid.

RECOMMENDED ACTION:
Perform an immediate inspection of the pump to identify the source and extent of leakage, with urgent repair or replacement if necessary.

Regulatory Reference

ISO 4413 - Hydraulic fluid power: General rules and safety requirements

API 686 - Recommended Practice for Machinery Installation and Installation Design

ISO 18435 - Diagnostics and prognostics for hydraulic systems

All maintenance actions should be documented in the facility CMMS and approved by the maintenance superintendent before execution.